

2(a)	any three from: good / universal, solvent or (is polar so) can dissolve (named) polar molecules / ions ; R organic solvent transport, qualified ; high specific heat capacity ; stabilises / regulates / maintains, constant / AW, (blood) temperature ; AVP ; e.g. water is, liquid at body temperature / a fluid, so flows	3
2(b)	any two from: as, hydrogencarbonate (ions) / HCO_3^- ; A bicarbonate for hydrogencarbonate <u>dissolved</u> in plasma ; bound to haemoglobin / forms carbaminohaemoglobin ; AVP ; e.g. as carbonic acid <u>in plasma</u>	2

Question	Answer	Marks
2(c)(i)	<i>any four from:</i> <i>similarities:</i> globular (proteins) ; contain / AW, prosthetic group / conjugated protein / porphyrin ring / haem / iron / Fe²⁺ ; contain / AW, alpha / α, helices ; both have a tertiary structure ; <i>differences:</i> myoglobin has one polypeptide and haemoglobin has four, polypeptides / subunits ; myoglobin has no quaternary structure and haemoglobin has quaternary structure ; myoglobin has one haem and haemoglobin has four haem (groups) ; A prosthetic group(s) for haem AVP ;	4
2(c)(ii)	<i>any two from:</i> (efficient / fast / AW) transfer of oxygen from haemoglobin to myoglobin ; to allow muscle cells to, respire more quickly / produce more ATP / respire aerobically (for longer) ; AVP ; ; e.g. idea of oxygen stored in myoglobin reduces likelihood of building up oxygen debt maintain (steep) concentration gradient (from blood to muscle cells)	2